In-Class Activity: Bioavailability

Module 7a - Bioavailability and Bioequivalence · In-class activity · 30 Sep 2026

Name _____ Date _____

1a After oral administration of a single 500 mg tablet of an antihypertensive drug to a normal volunteer, the calculated AUC was 70 mg h/L. Following a 500 mg IV bolus dose of the same drug to the same volunteer, the elimination half-life was 3 hours and the apparent volume of distribution was 25 L. What is the absolute bioavailability of this oral tablet in this volunteer?

1b A drug has an elimination half-life of 3 hours and an apparent volume of distribution of 25 L; a 500 mg IV bolus dose gives an AUC of 86.58 (mg/L)hr. What is the expected AUC after administration of a single IV bolus dose of 250 mg to the same volunteer?

3a The equation for concentration of drug in the plasma as a function of time following a 500 mg IV bolus dose was found to be $C_p = 50e^{-0.2t}$. What is the clearance of the drug?

3b A 500 mg IV bolus dose of a drug gives $C_p = 50e^{-0.2t}$, and the clearance of the drug is 2 L/hr. What is the AUC of the drug following this 500 mg IV bolus dose?

3c A 500 mg IV bolus dose of a drug gives an AUC of 250 (mg/L)hr. If the AUC following a 500-mg oral tablet dose was found to be 188 mcg hr/mL, what is the oral bioavailability of the drug from the tablet?